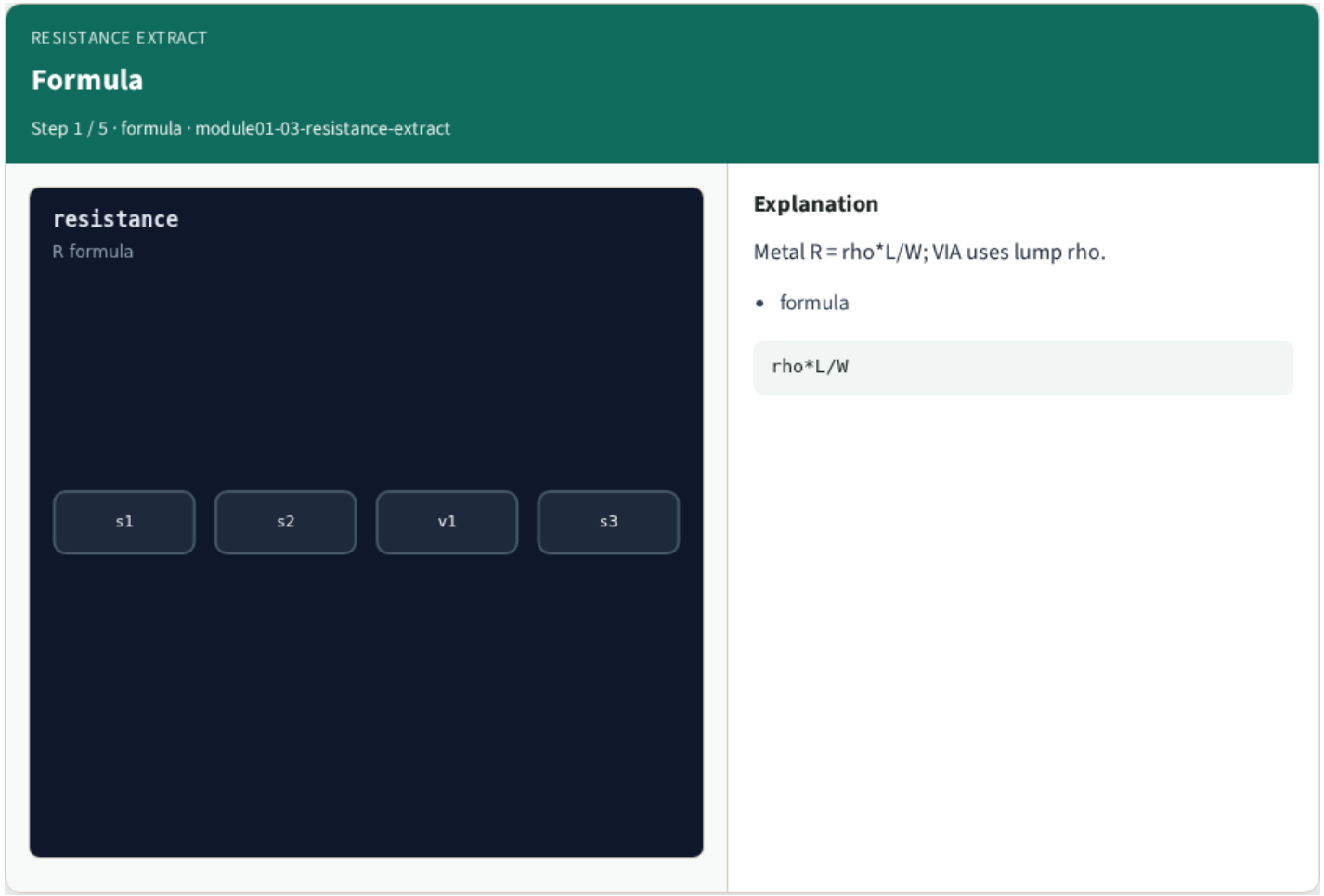
Resistance

Resistance extract computes R = rho*L/W for metals; VIA uses lump rho

The idea

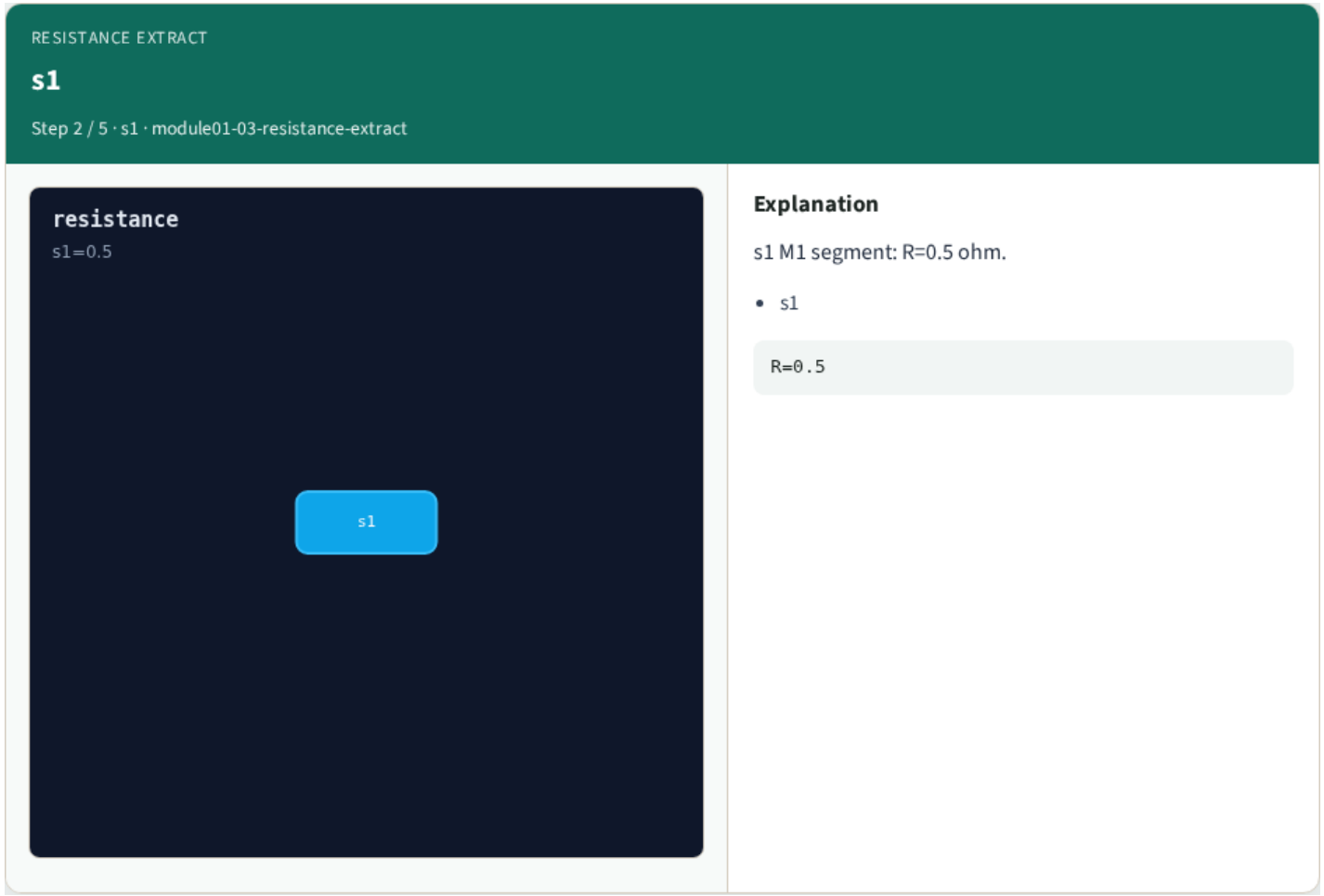
- GOLDEN: R_total=3.06; s1=0.5

Formula



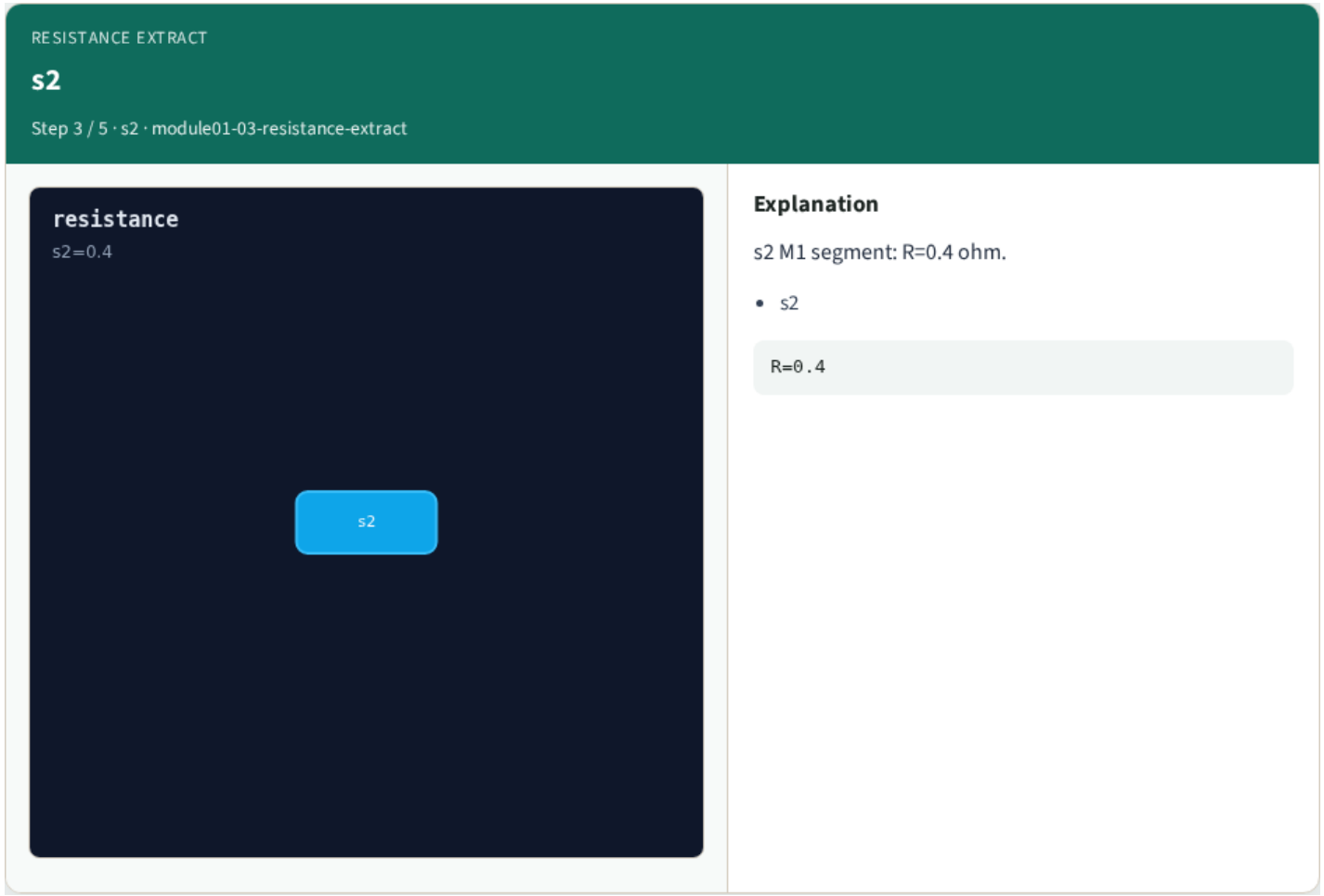
Formula

s1


RESISTANCE EXTRACT
s1
Step 2 / 5 · s1 · module01-03-resistance-extract
resistance
s1=0.5
s1
Explanation
s1 M1 segment: R=0.5 ohm.
s1
R=0.5

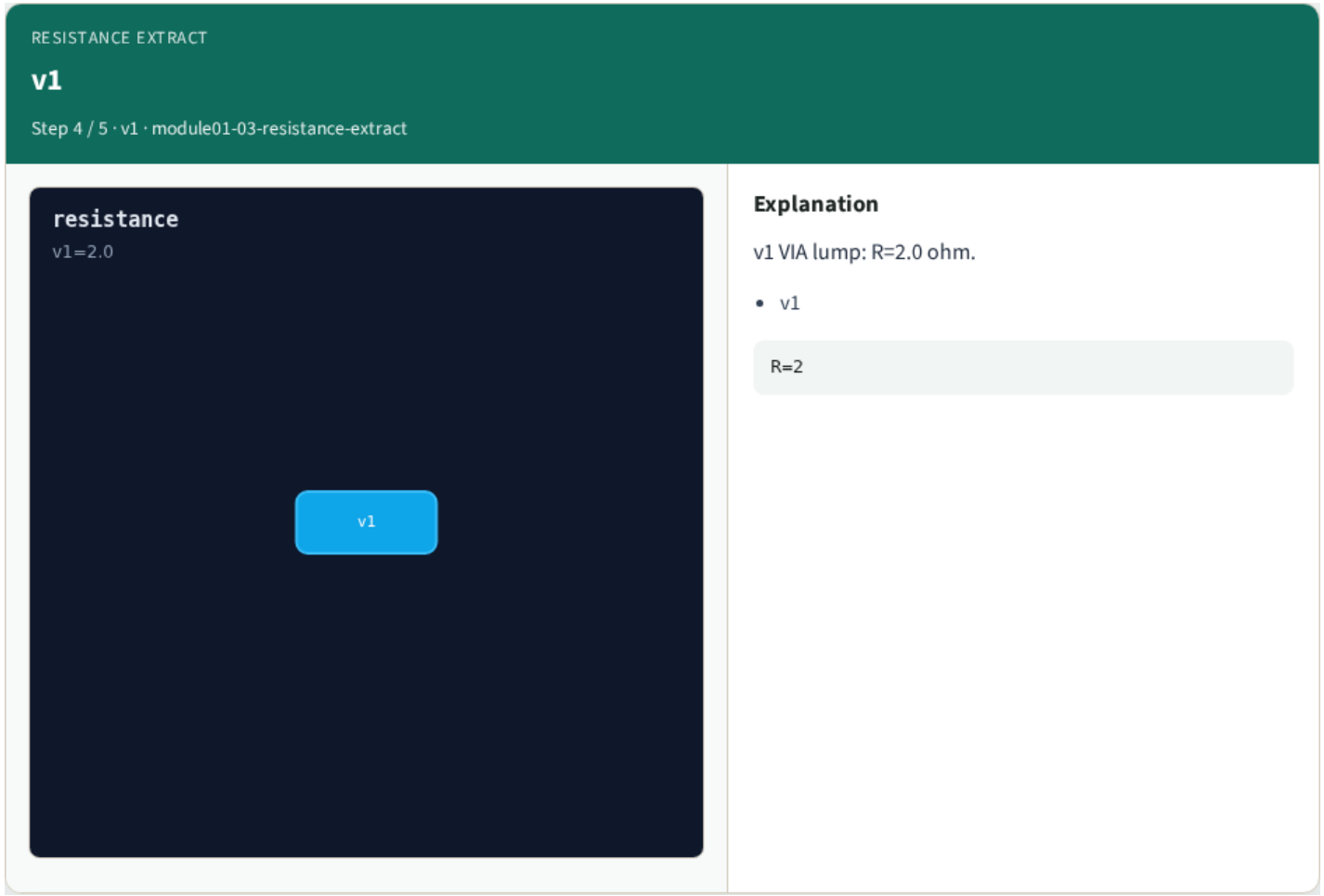
s1

s2


RESISTANCE EXTRACT
s2
Step 3 / 5 · s2 · module01-03-resistance-extract
resistance
s2=0.4
s2
Explanation
s2 M1 segment: R=0.4 ohm.
s2
R=0.4

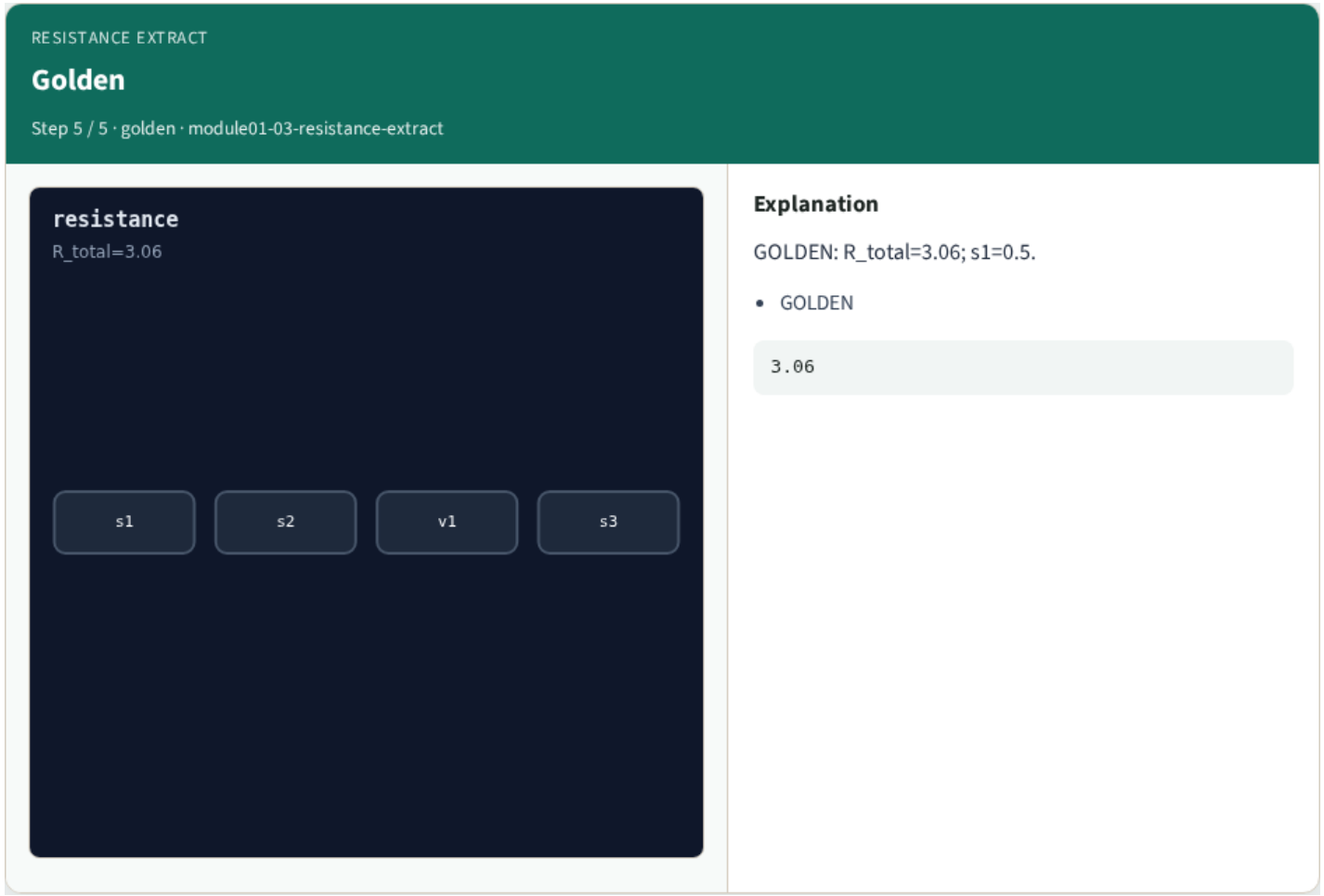
s2

v1


RESISTANCE EXTRACT
v1
Step 4 / 5 · v1 · module01-03-resistance-extract
resistance
v1=2.0
v1
Explanation
v1 VIA lump: R=2.0 ohm.
v1
R=2

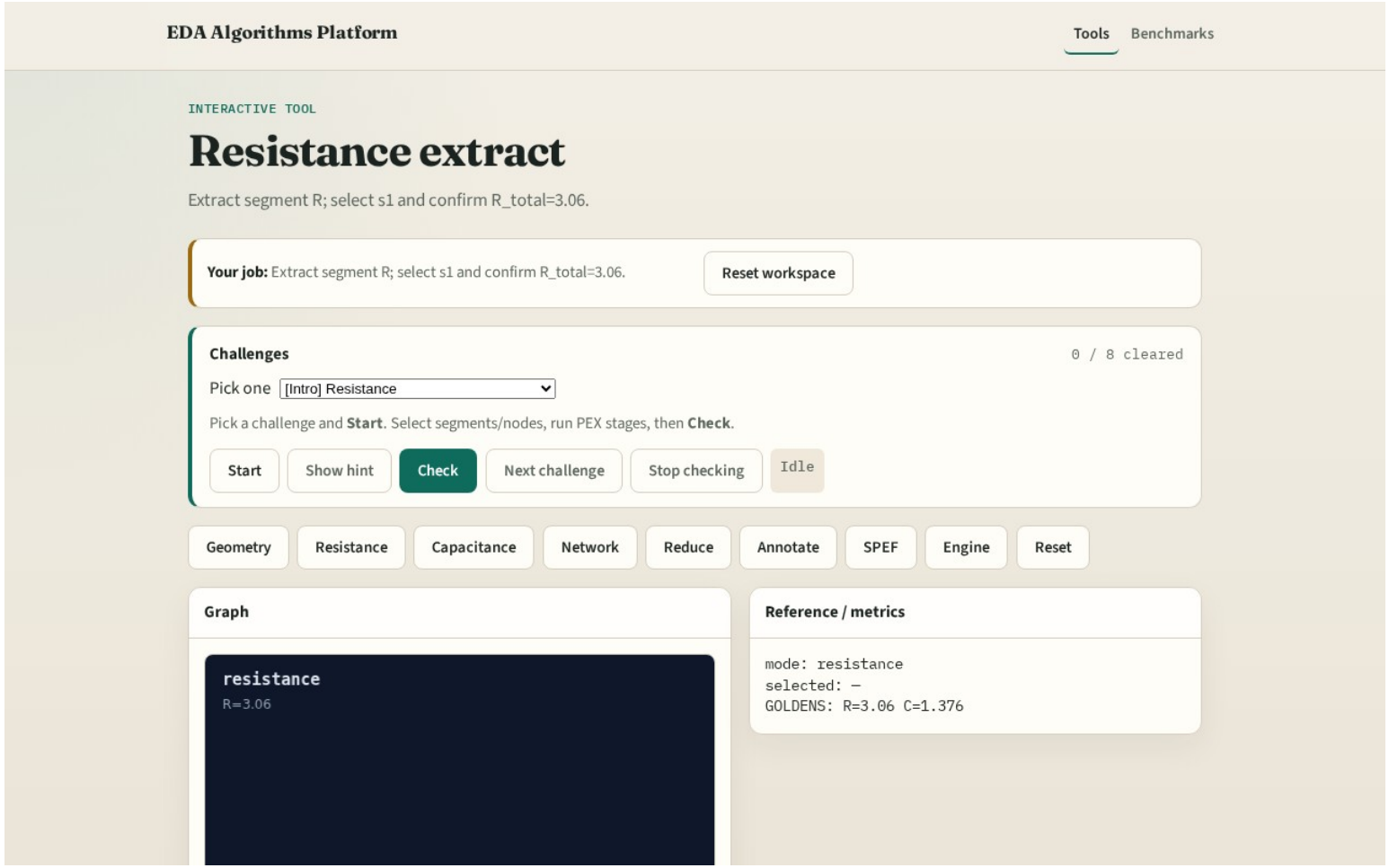
v1

Golden



Golden

Browser lab track



Browser lab starter

Implement track

- Call extract_resistance(); assert r_total==3.06 and resistors['s1']==0.5

Pitfalls

- Using max edge as width
- Forgetting VIA lump resistance

Your turn

- Next: capacitance extract